TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*A technology that allows the spatial-resolved defect measurements of SOEC stacks and cross-scale degradation analysis.*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Electrics and Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	New/renewable energy	Hydrogen	Measuring instruments	Physical Quantity Test/Analysis Instrument
Concept of Proposed Technology	The proposal develops and validates a reliability-oriented diagnostic framework for ESC-based solid oxide cells and short stacks. It links microstructural and interfacial heterogeneities to local current distribution, thermal behavior, and degradation mechanisms under industrial operating conditions.			
Rationale for Joint R&D Project	Collaboration combines Fraunhofer's characterization expertise with Korean SMEs' aims to scale ESC-based SOEC/SOFC from laboratory cells to short stacks with reliable performance. Joint R&D reduces uncertainty in failure modes, accelerates durability improvements, and establishes validated protocols.			
Key Words	1. Cross-scale Diagnostics of ESC-based Cells and Stacks 2. Spatially Resolved Electrochemical & Thermal Analysis 3. Root-cause of Degradation Mechanism 4. Reliability-Oriented Diagnostic Framework for Scale-up			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	-Identify and quantify dominant degradation mechanisms tied to microstructural and interfacial heterogeneities in ESC-based SOEC/SOFC cells and short stacks. - Map local current density and thermal inhomogeneities under industrial operating conditions; detect initiation points during accelerated stress testing. - Differentiate material-driven versus operation-driven degradation and establish cause–effect relationships between microstructure, operation, and lifetime. - Develop reliability-oriented diagnostic protocols and operating windows suitable for industrial validation and customer qualification.			
R&D Target Specifications	-Cross-scale diagnostic coverage: atomic/interface to ESC and short-stack level. -Spatial resolution and sensitivity sufficient to detect critical interfaces, pores, cracks, delamination, and non-uniform porosity/density. - Operating envelopes covering high current density, elevated temperature, dynamic load, and thermal cycling with defined acceptance criteria (accuracy, repeatability, reproducibility). - Validated protocols and metrics enabling correlation to durability (lifetime, degradation rates) and risk mitigation during scale-up			
R&D Plan	Phase I – Pre-study: • Risk identification and feasibility validation. • Map critical interfaces and microstructural risk factors; define diagnostic parameters and test protocols. • Identify dominant degradation mechanisms; set go/no-go criteria for scale-up strategies. Phase II – Main R&D: -Implement cross-scale diagnostics and spatially resolved electrochemical/thermal analysis under industrially relevant conditions. -Conduct root-cause investigations under high current density, elevated temperature, dynamic load, and thermal cycling. -Optimize cell (and if budget allows, short-stack) designs based on diagnostic insights; validate durability improvements. - Deliver a diagnostic toolbox, operating windows, and design guidelines; validate in a production-like environment and prepare for customer-oriented reliability testing.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	-microstructural defect analysis on reference and stressed SOEC cells - Electrochemical evaluation of single cells and short stacks (accelerated stress tests, EIS etc.) - Root-cause analysis of degradation mechanism after SOEC operation - Testing and verification of thermal behaviour and long-term durability - Mapping of critical interfaces and microstructural risk factors - Definition of key diagnostic parameters and test protocols
Expected R&R of Korean SME Partner	○ Synthesize SOEC cells that are ESC-based (electrolyte-supported cell) ○ Build short-SOEC stacks for durability evaluation of single cells and 3-cells ○ Inspect cross-section of the SOEC cells before and after the operation ○ Analyze root-cause of failure modes during SOEC operation on single cell and stack levels. ○ Commercialize the cells and stacks, Clear go/no-go criteria for scale-up strategies

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	ㅇunderstanding the degradation mechanisms occurring during accelerated aging tests ㅇhigh-resolution microstructural investigations on reference and stressed cells/stacks to elucidate the causes of degradation mechanisms diagnostically. ㅇmaterial validation and optimization of operating conditions for newly developed stack designs • Scientific understanding of degradation mechanisms in ESC-based SOEC technology • Practical diagnostic framework enabling reliable scale-up • Reduced technical risk for industrial deployment • Strengthened competitiveness of Korean SME in hydrogen and energy markets Establishment of a strong Korea–Germany collaboration model for industrial electrochemical technologies

6. References

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(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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